

Algebra Foundations: Shadow Skills Check-up

1. Collecting Like Terms

Simplify the following expressions:

- a) $5a + 2b - 14a + 7b$
- b) $3x^2y - 6xy + 4xy^2 - 8x^2y$

2. Multiply with Powers

Simplify the following:

- a) $4p^2q \times 3p^3q^4$
- b) $2a^3b^2c^5 \times 6ab^4c^3$

3. Divide with Powers

Simplify:

- $\frac{6x^5y^3z^2}{18x^2y^7z^2}$

4. Powers of Powers

Simplify:

- $(4a^2b^5)^3$

5. Substitution

Find the value of W when $x = 15$ and $y = -4$:

- $W = \frac{3x-5y}{5}$

6. Rearrange Formula

Make x the subject of the formula:

- $A = 5px$

7. Expanding Brackets

Expand the following:

- Single Bracket: $3y(4x^2 - 5y^3)$
- Quadratics: $(x - 4)(x + 7)$
- Quadratics: $(2x + 3)(5 - 2x)$

8. Factorising

Factorise completely:

- Single Bracket: $24a^2b - 12ab^2 + 6ab$

- Quadratics: $x^2 + 10x + 21$
- Quadratics: $x^2 - 3x - 28$

9. Special Case Factorisation

Factorise:

- Perfect Square: $x^2 - 12x + 36$
- Difference of Two Squares: $x^2 - 49$

10. Non-Unit Coefficient Factorisation

Factorise:

- $2x^2 + 14x + 20$
- $3x^2 + 7x - 6$

11. Algebraic Fractions

Perform the operations and simplify:

- Multiplication: $\frac{4x-8}{3} \times \frac{x+2}{8}$
- Addition: $\frac{x-3}{4} + \frac{x+5}{3}$

12. Solving Linear Equations

Solve for x :

- One/Two Step: $15x = 75$
- One/Two Step: $\frac{x-10}{6} = 4$
- x on both sides: $14x - 9 = 6x + 23$
- Fractional: $\frac{x+3}{x-5} = \frac{x-2}{x-8}$

13. Linear Inequalities

Solve the inequality:

- $-3x - 4 + 10 < -9$

14. Simultaneous Equations

Solve to find x and y :

- By Elimination: $4x + 3y = 18$
 $2x - 5y = 22$
- By Substitution: $y = 12 - 3x$
 $5x - 2y = 31$

End of Skills Check-up